

Volume I. Introduction.

Beginning of the literal backfill.

We presuppose, in our considerations, the natural numbers $1, 2, 3, \dots$ and the rules according to which calculations are made with these numbers as known and given. The fundamental arithmetical operations, the so-called four species, are addition, multiplication, to which, as repetition, exponentiation belongs, subtraction, and division. The first two are called the direct operations of calculation; they are distinguished by the fact that in the realm of the natural numbers they can be carried out without restriction. The first of the indirect or inverse operations, subtraction, can be carried out only when the minuend is greater than the subtrahend.

The problem of division can be understood in two ways. In the first elementary interpretation one asks how often the divisor is contained in the dividend. A number is not contained in a smaller one. But if the dividend is equal to or greater than the divisor, then the answer to the question gives a quotient and, in most cases, a remainder which is smaller than the divisor. If no remainder is left, one says that the division comes out exactly, or that the dividend is divisible by the divisor, or that the divisor is a factor or divisor of the number which forms the dividend.

This problem, which already presents itself at the first stages of the art of calculation, leads to a more deeply lying distinction among numbers, which is the foundation of all number theory.

Since a factor can never be greater than the number of which it is a factor, every number has only a finite number of factors. Every number is divisible by 1 and by itself, and a number which has no other divisor is called a prime number. The number 1 itself is customarily not designated a prime number, for reasons of expediency. If two numbers are divisible by a third, then the sum and the difference of the first two are also divisible by the third; and if one number is divisible by a second and this second by a third, then the first is also divisible by the third.

Two numbers always have the common divisor 1. If they have no other common divisor, as for example the pairs of numbers 5 and 7, or 21 and 38, then the two numbers are called relative prime numbers, or also numbers prime to one another.

Among the common divisors of any two given numbers one will be the greatest, and it is a very important problem to find this greatest common divisor. This is accomplished by a procedure known under the name algorithm of the greatest common divisor, which already occurs in Euclid.¹

Let a, a_1 be the two numbers; take the larger of them, say a , as dividend and the smaller a_1 as divisor, and determine the quotient q_1 and, if the division does not come out exactly, the remainder a_2 , so that

$$a = q_1 a_1 + a_2,$$

and $a_2 < a_1$. Every common divisor of a and a_1 is then also a common divisor of a_1 and a_2 , and conversely. If one proceeds with a_1, a_2 just as with a, a_1 , and puts, when the division does not come out exactly,

$$a_1 = q_2 a_2 + a_3,$$

then again $a_3 < a_2$, and every common divisor of a_1 and a_2 is also a common divisor of a_2 and a_3 , and conversely. If one continues dividing in this way, then, since the numbers $a, a_1, a_2, a_3, \dots$ continually decrease, the division must necessarily come out exactly after a finite number of steps; that is, at last a pair of equations

$$a_{\nu-2} = q_{\nu-1} a_{\nu-1} + a_{\nu}, \quad a_{\nu-1} = q_{\nu} a_{\nu}$$

must occur. Then a_{ν} is a common divisor of all preceding a 's, hence also of a and a_1 , and every common divisor of a and a_1 is a divisor of a_{ν} . Thus a_{ν} is the greatest common divisor of a and a_1 , and at the same time we have arrived at the theorem that every common divisor of two numbers must divide their greatest common divisor.

¹Elements, Book VII, 11, vol. II of Heiberg's edition.

If a and a_1 are relatively prime numbers, then the last divisor is $a_\nu = 1$. Multiplying, under this assumption, the preceding equations by an arbitrary number b , it follows that b is the greatest common divisor of ab and a_1b , and hence that every common divisor of ab and a_1 , or of a and a_1b , must be a divisor of b . Thus if a_1 is relatively prime to a and to b , it is also relatively prime to ab ; and from the special assumption that a_1 is a prime number it follows that a product can be divisible by a prime number only when at least one of the factors is divisible by it. Thus, if the product ab is divisible by a_1 and a_1 is relatively prime to a , then b must be divisible by a_1 .

Let a, b be any two numbers with greatest common divisor d , and let $a = da'$, $b = db'$. Then a' and b' are relatively prime to one another. Every number m which is at once a multiple of a and of b has the form $m = am' = da'm'$, and in this $a'm'$ must be a multiple of b' ; therefore m' too must be a multiple of b' . That is, every number which is at once a multiple of a and of b is divisible by $a'b'd$. This number $a'b'd$, which itself is a common multiple of a and b , is therefore called the least common multiple of a and b .

If a number is divisible by two relatively prime numbers, then it is also divisible by their product; and if, therefore, a number m is divisible by several numbers of which each two are relatively prime to one another, then it is also divisible by the product of all these numbers.

On this rests the proof of the important theorem that a number m can always be represented, and in only one way, as a product of prime numbers. For since a divisor cannot be greater than the dividend, m can certainly be divisible by only a finite number of prime numbers. If a is one of these prime numbers and a^α the highest power of a which divides m , then α too is a definite number. Let likewise $b^\beta, c^\gamma, \dots$ be the highest powers of the remaining prime numbers $b, c, \dots$ which divide m . Then, since any two of the numbers $a^\alpha, b^\beta, c^\gamma, \dots$ are relatively prime, m must also be divisible by the product $a^\alpha b^\beta c^\gamma \dots$; and it must be equal to this product, since otherwise another prime number, or a higher power of one of the prime numbers $a, b, c, \dots$, would have to divide m .

We now give an overview of the extensions of the concept of number which are necessary in mathematics and which were gradually introduced.

By a manifold or set, or by the abbreviated sign M , we understand a system of objects or elements of any kind, which is so delimited and complete in itself that, for every arbitrary object, it is completely determined whether it belongs to the system or not, regardless of whether we are able, in each particular case, actually to make the decision or not.

A set is called ordered when, of any two different elements, one that is fully determined in itself always counts as the greater, and in such a way that from $a > b$, $b > c$ one always has $a > c$. If $a > b$ and $b > c$, or $a < b$ and $b < c$, we say that b lies between a and c .

The natural numbers form an ordered set; between two successive elements of it there lies no further element. Such a manifold is called discrete. An ordered set with the property that between any two elements further elements can always be found is called dense. One can form a dense set by combining the natural numbers in pairs and regarding these pairs as elements of a set. These pairs shall be called fractions and denoted by $m : n$ or m/n , and two such fractions $m : n$ and $m' : n'$ are set equal when $mn' = nm'$. If all fractions equal to one another are combined into one element, one obtains a manifold which is ordered once one further stipulates that $m : n$ is greater than $m' : n'$ when $mn' > nm'$. That this manifold is dense is seen as follows: if $\mu = m : n$, $\mu' = m' : n'$ are two fractions and $\mu > \mu'$, then, with h arbitrary, one can put

$$\frac{hmn'}{hnn'}, \quad \frac{hm'n}{hnn'},$$

and therein $hmn' > hm'n$. One can always take h so large that further numbers lie between hmn' and $hm'n$, and if p is such a number, then $p : hnn'$ lies between μ and μ' .

The points of a straight line can also be regarded as an ordered set if by greater and smaller one understands some indication of position, for instance farther right and farther left, or higher and lower.

A division of an ordered set M into two parts A, B of such a kind that every element a of A is smaller than every element b of B is called a cut in M and is suitably denoted by (A, B) . Such a

cut arises when one selects an element μ of M , assigns all smaller elements to A , all larger elements to B , and assigns μ itself at will to A or to B . Strictly speaking, according as one does the one or the other, two cuts arise, but we shall always regard them as equal. If, in a cut (A, B) , either A contains a greatest element or B a least element μ , then we say that μ produces the cut (A, B) . But it may also occur that neither A has a greatest nor B a least element.

If every cut in a dense set is produced by a definite element μ , then the set is called continuous.

Continuity as well as density are properties which, by the nature of the matter, are inaccessible to our sense perception; consequently they also cannot be demonstrated with rigor in things of the external world, in spatial magnitudes, intervals of time, or masses, however much they may seem to lie in the essence of our intuition. But pure conceptual systems can very well be constructed to which density without continuity, or also density and continuity, belong.²

²This was proved by Dedekind, to whom we owe in general the definition of continuity given above. Compare Dedekind's writings, *Stetigkeit und irrationale Zahlen*, Braunschweig 1872, 1892, and *Was sind und was sollen die Zahlen?*, Braunschweig 1888, 1893. Other manifolds to which continuity belongs were formed by Weierstrass and G. Cantor.